Instructor Key — Exercise 8: Plates, Peak Shape and the Sample's Own Conductivity

Every value below was measured against drylab v0.24.7 with the app's own readout logic (`validation/ex8/harness.py` reproduces the CE cell: plug at 0.2 L with a 1/e half-width of 0.8 mm, `detector_trace`, FWHM from the saved times, apparent mobility from plug center to detector node). `validation/ex8/measure_ex8.py` regenerates `validation/ex8/key_values.json` in four sections of under a minute each. Section: **CE**. Grid points **257** throughout, as the sheet instructs. No CSV export exists in the CE section; every number is on screen.

Hand values versus the app. Plate counts below are computed the way students compute them, from the *displayed* (rounded) migration time and FWHM. FWHM is quantized to the saved time step (0.039 s at 500 V and 257 points), so expect about ±5 % scatter between groups.

Known app defect students will meet (v0.24.7). Time slice 0 carries the solver's hydronium *starting guess*, not the equilibrium, so pH and conductivity at slice 0 read wrong (pH 7.00 and 0.113 S/m in the Tris BGE, against the true 4.93 and 0.0836 S/m). Every slice from 1 (t ≈ 0.03 s) onward is correct, and the run itself is unaffected. The sheet sends students to slice 1. Logged in `.context/memory/issues.md` with the one-line fix.

Timing. Nine runs in class, 7 to 20 s each at 257 points.

Getting started

Tris BGE, samples at 0.1 mM, 500 V, 20 s, detector 15 mm, EOF 0.

Quantity	App shows
Field E	16667 V/m
Current density j	1394.1 A/m²
Joule heating	0.05 W/m (radial dT 0.01 K)
Background pH	4.93

Analyte	t (s)	FWHM (s)	N (hand)
potassium K+	7.22	1.13	226
sodium Na+	10.52	1.59	243
lithium Li+	13.59	2.02	251

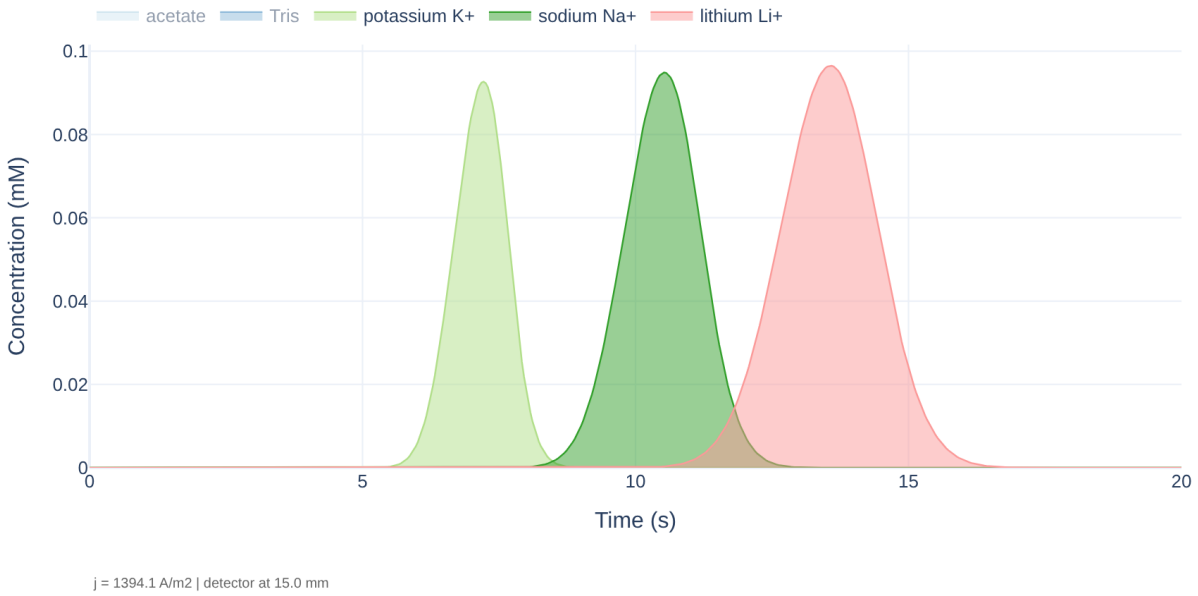


Figure 1. Electropherogram students should see at the start (Tris BGE, 0.1 mM samples, 500 V, detector 15 mm). Three symmetric peaks. BGE traces are greyed out in the legend because the app hides them.

Q1 — Conductivity from the current

$$\kappa = \frac{j}{E} = \frac{1394.1\ \mathrm{A\,m^{-2}}}{16667\ \mathrm{V\,m^{-1}}} = 0.0836\ \mathrm{S\,m^{-1}}$$

The overlay at slice 1 reads **0.0836 S/m** away from the plug (0.0864 S/m at the plug center, where the 0.1 mM sample cations add a little). They agree because the app sets j once from the BGE conductivity and the field, j = κE. **The point to land:** the voltage is the control, and the current is a property of the buffer. A group that reads 0.113 S/m has read slice 0 (see the defect note).

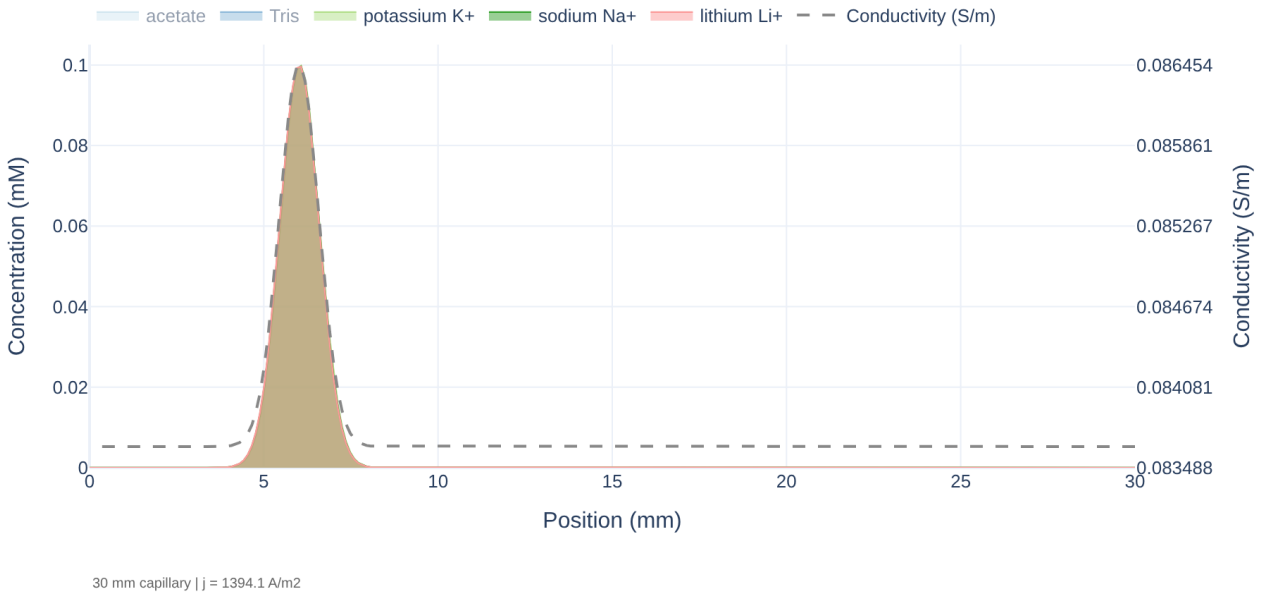


Figure 2. Spatial profile at time slice 1 with the conductivity overlay (dashed, right axis). The BGE reads 0.0836 S/m. The plug at 6 mm reads slightly higher. Plotly auto-scales the right axis to the tiny 3 % bump, so its tick labels look odd; students should read the flat baseline.

Q2 — Diffusion-only prediction

$$N = \frac{zV\ell}{2(k_B T/e)L} = \frac{(1)(500)(9)}{2(0.02569)(30)} = 2920 \qquad N_{1500} = 8760$$

Ratio **3**: N should grow in proportion to V.

Q3 — Measured

Voltage (V)	t K⁺ (s)	FWHM K⁺ (s)	N K⁺	Joule (W/m)
500	7.22	1.13	226	0.05
1000	3.61	0.54	248	0.18
1500	2.41	0.35	263	0.41

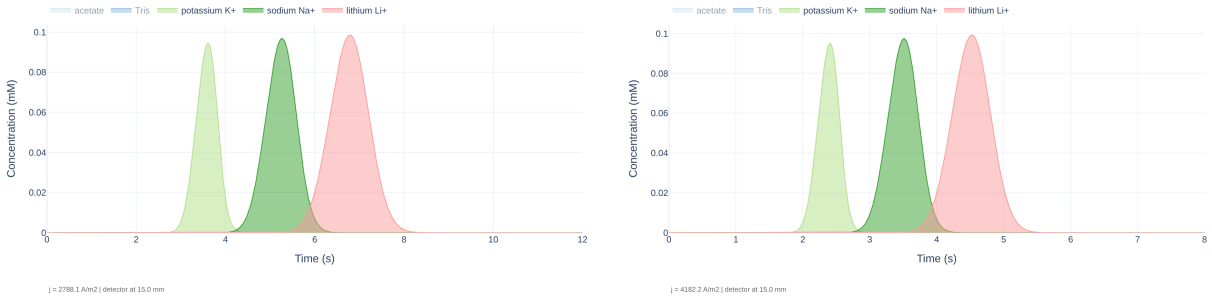


Figure 3. 1000 V (left, 12 s run) and 1500 V (right, 8 s run). Peaks arrive sooner and look narrower in seconds, but N barely moves. Note j in the stamp doubles and triples with the voltage (2788.1 and 4182.2 A/m²): the current follows the field.

N rises only **16 %** for a threefold voltage, and is about 1/13 of the diffusion-only prediction at 500 V.

Q4 — Reconcile: the injection dominates

$$D = \mu_{ep}\frac{k_B T}{e} = (76.2\times10^{-9})(0.02569) = 1.96\times10^{-9}\ \mathrm{m^2\,s^{-1}}$$

$$\sigma^2_{diff} = 2Dt = 2(1.96\times10^{-9})(7.22) = 2.83\times10^{-8}\ \mathrm{m^2} = 0.0283\ \mathrm{mm^2}$$

$$\sigma^2_{inj} = (0.57)^2 = 0.325\ \mathrm{mm^2}$$

The injection variance is **11.5 times** the diffusion variance.

$$N = \frac{\ell^2}{\sigma^2_{inj}+\sigma^2_{diff}} = \frac{81}{0.325+0.0283} = 229$$

Measured 226. **Why voltage barely helps:** raising V shortens t and shrinks σ_diff², but σ_inj² does not depend on V and already makes up 92 % of the total. The small rise from 226 to 263 is the diffusion term shrinking toward zero. The limit is $\ell^2/\sigma_{inj}^2 \approx 250$ (253 with the exact plug σ of 0.566 mm), and 263 sits on it within the ±5 % FWHM-reading scatter.

Watch for groups that blame Joule heating. The readout answers that: 0.41 W/m and a radial ΔT of 0.05 K at 1500 V. Heating would lower N as V rises, and N went up.

Q5 — Detector at 25 mm, predicted

ℓ = 25 − 6 = **19 mm**. Expected t ≈ 7.22 × 19/9 = 15.2 s, so

$$\sigma^2_{diff} = 2(1.96\times10^{-9})(15.2) = 0.0596\ \mathrm{mm^2} \qquad N = \frac{361}{0.325+0.0596} = 939$$

FWHM in seconds: **about the same**. The injection width in time is σ_inj / v, and v is unchanged at fixed V.

Q6 — Measured

Detector (mm)	t K⁺ (s)	FWHM K⁺ (s)	N K⁺
25	15.19	1.13	1001

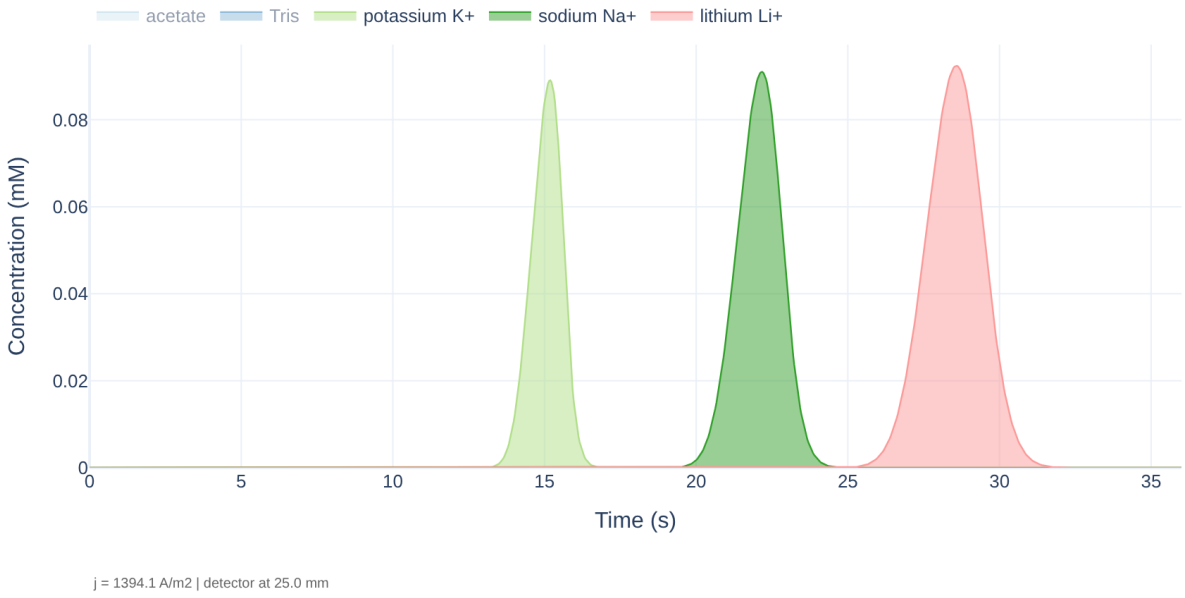


Figure 4. Detector at 25 mm (500 V, 36 s run). Same peak widths in seconds as Figure 1, but spread over a longer time axis: the peaks look sharper because N is 4.4 times larger.

N rose by **4.4 times** (226 → 1001), close to (19/9)² = 4.46. The width in seconds did not change (1.13 s). **The lever when the injection dominates is migration distance (or a shorter plug), not voltage.** The zone keeps the same physical width while the distance it is measured against grows.

Part B baseline — imidazole co-ion, 0.1 mM

Analyte	t (s)	FWHM (s)	Shape
K⁺	7.13	1.09	symmetric
Na⁺	10.43	1.55	symmetric
Li⁺	13.45	2.06	symmetric

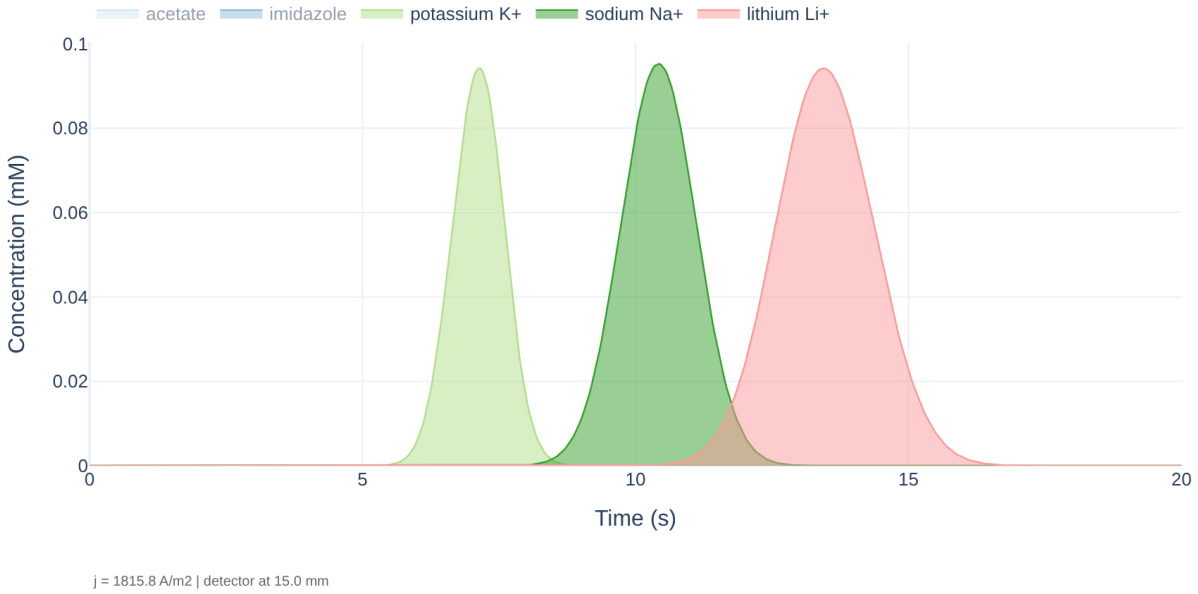


Figure 5. Imidazole co-ion, 0.1 mM samples. Indistinguishable from Figure 1 at this concentration.

pH stays 4.93. Imidazole (pKa 7.05) is fully protonated at this pH, so 12 mM imidazolium balances 12 mM acetate, exactly as Tris did.

Q7 — Predictions

Analyte	vs imidazolium (52.0)	Shape	t at maximum
K^+ (76.2)	faster	diffuse front, sharp rear	later
Na^+ (51.9)	matched	symmetric	about the same
Li^+ (40.1)	slower	sharp front, diffuse rear	earlier

Q8 — Measured at 2 mM

Analyte	t (s)	FWHM (s)	Height (mM)	Asymmetry at 10 % (rear/front)
K^+	7.99	1.58	1.421	0.37
Na^+	10.78	1.77	1.705	1.00
Li^+	13.27	2.79	1.416	1.86

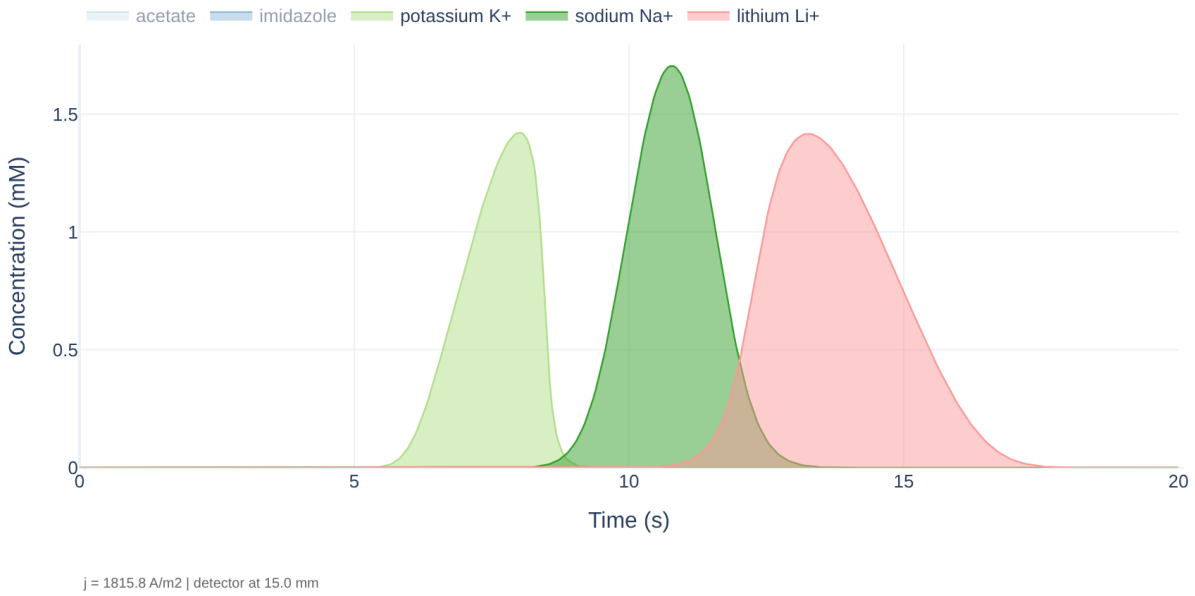


Figure 6. Imidazole co-ion, 2 mM samples. K^+ (faster than the co-ion) has a slow rise and a sharp drop. Na^+ (matched) stays symmetric. Li^+ (slower) has a sharp rise and a long tail. Remind students that in a time trace the front of the zone arrives first, so a diffuse front is the slow rise on the left.

The asymmetry column is for the instructor: the app does not report it, and students describe the shapes in words. K^+'s maximum moved later (7.13 → 7.99 s) and Li^+'s moved earlier (13.45 → 13.27 s), as predicted.

Q9 — Reconcile

K^+ displaces imidazolium to keep the zone electroneutral. K^+ is faster, so the zone **conducts better**. The current density is the same everywhere, so the field inside the zone is **lower** ($E = j/\kappa$). A K^+ ion that diffuses out ahead enters the higher field of the BGE and runs further ahead, so the front spreads. One that falls behind also enters the higher field and runs back into the zone, so the rear stays sharp. Na^+ has the same mobility as imidazolium, so replacing one with the other leaves κ, and therefore the field, unchanged.

Honest note for discussion: Na^+ still broadened slightly (1.55 → 1.77 s, N 251 → 205) while staying symmetric. All three ions start in one plug, and during the first seconds the K^+ and Li^+ zones overlap the Na^+ zone, changing its local field. Strong groups notice this. It is real in the model and does not contradict the rule.

Q10 — Salty sample, predicted

Wider and shorter. The salt raises the plug conductivity, so the field in the plug is lower. K^+ at the front of the plug crosses into the BGE and speeds up, while K^+ still inside lags, and the zone stretches as it leaves.

Q11 — Measured (30 mM NaCl)

Quantity	Value
K^+ t	8.59 s
K^+ FWHM	4.26 s
K^+ height	0.027 mM
κ in the plug (slice 1)	0.488 S/m
κ in the BGE (slice 1)	0.109 S/m

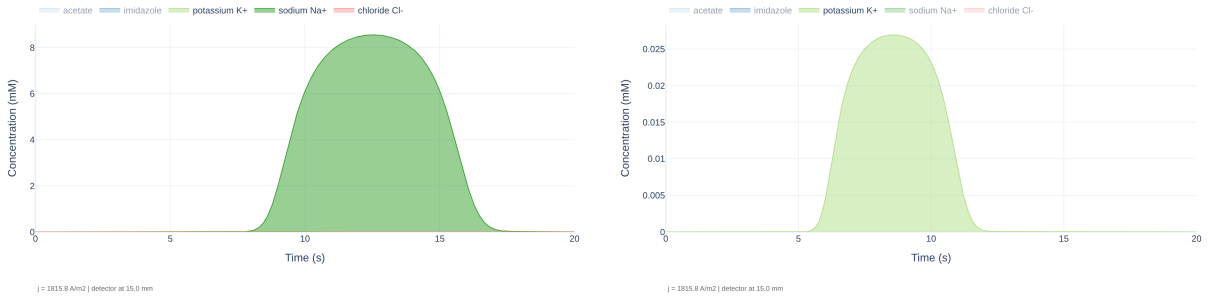


Figure 7. Salty sample as first drawn (left): the 30 mM sodium peak fills the axis and K^+ is invisible. After clicking sodium Na+ and chloride Cl- off in the legend (right), the K^+ zone appears flat-topped and about four times wider than in Figure 5. The faint chloride rise at the far right of the left panel is numerical residue at the 0.001 mM level.

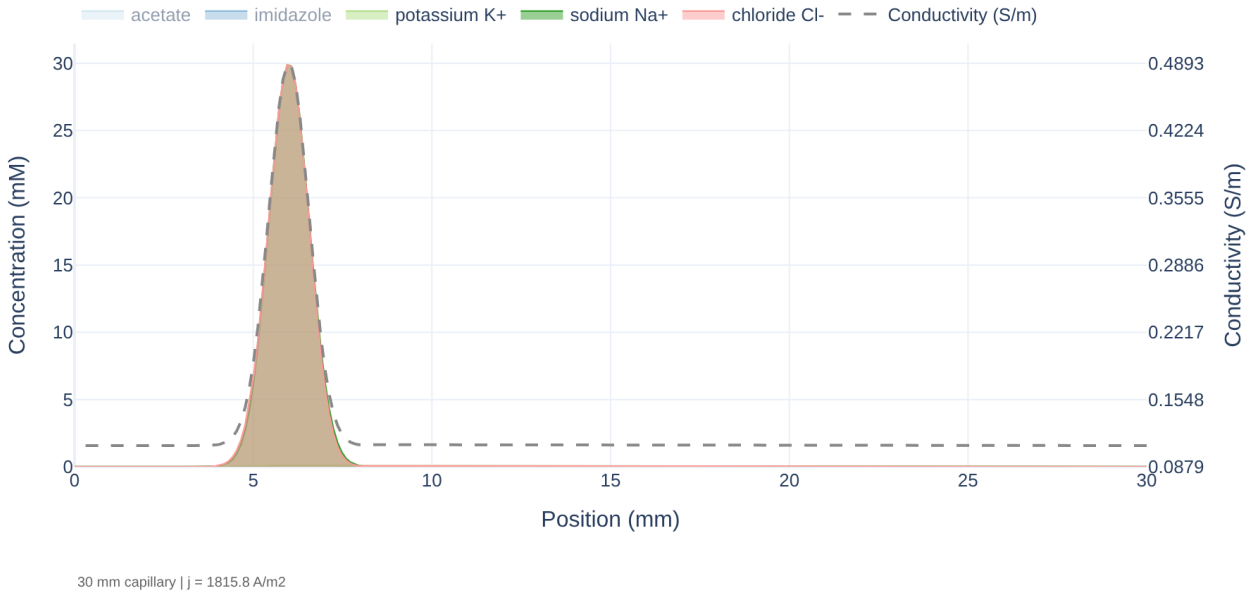


Figure 8. Spatial profile at time slice 1 with the conductivity overlay: 0.488 S/m at the plug center against 0.109 S/m in the BGE.

Q12 — Reconcile

$$\frac{\text{FWHM}_{\text{salty}}}{\text{FWHM}_{\text{B}}} = \frac{4.26}{1.09} = 3.9 \qquad \frac{\kappa_{sample}}{\kappa_{BGE}} = \frac{0.488}{0.109} = 4.5$$

The zone stretched by close to the conductivity ratio, as ℓ_zone ≈ ℓ_inj κ_sample/κ_BGE predicts. The equation treats the plug boundary as fixed. In the run the salt zone itself migrates and spreads, which is why the agreement is close but not exact. The effect is **antistacking** (also called destacking). The peak height fell to 29 % of the Part B value, and the migration time grew from 7.13 to 8.59 s because K^+ spent its first seconds in the low-field plug.

On the warning: it is related but not the same mechanism. Both come from a local change in conductivity setting a local field. In Part B the *analyte's own* ions changed κ and carried that change with them for the whole run, so the peak became asymmetric. Here the *sample matrix* sets the field during injection and loading, and the K^+ peak stays nearly symmetric (asymmetry 1.04) but wide. The warning is right that the sodium zone itself is far too concentrated.

Fixes: dilute or desalt the sample, make it up in BGE or water, or raise the BGE concentration so κ_BGE approaches κ_sample.

Extension (if a group finishes early, 10 mM NaCl): FWHM 2.12 s, a ratio of 1.95 against a conductivity ratio of 0.236/0.109 = 2.17.

Part D — pK_a, pH and migration order

Imidazole BGE (12 mM, pH 4.93), 1000 V, 30 s. Sample: K^+ marker, pyridine and aniline, each 0.1 mM.

Q13 — Predictions

Fully protonated, **anilinium is faster** (32.5 against 30.0).

$$\alpha_{pyr} = \frac{1}{1+10^{4.93-5.16}} = 0.629 \qquad \alpha_{an} = \frac{1}{1+10^{4.93-4.80}} = 0.426$$

$$\mu_{eff,\,pyr} = 0.629 \times 30.0 = 18.9 \qquad \mu_{eff,\,an} = 0.426 \times 32.5 = 13.8 \quad (\times 10^{-9}\ \text{m}^2\,\text{V}^{-1}\,\text{s}^{-1})$$

$$t = \frac{\ell}{\mu_{eff} E}: \quad t_{pyr} = \frac{9 \times 10^{-3}}{(18.9 \times 10^{-9})(33333)} = 14.3\ \text{s} \qquad t_{an} = 19.5\ \text{s}$$

Predicted order: K^+, then **pyridine**, then aniline.

Q14 — Measured

Analyte	t (s)	Apparent mobility	Predicted t (s)
K^+	3.57	76.1	3.5
pyridine	13.95	19.5	14.3

Analyte	t (s)	Apparent mobility	Predicted t (s)
aniline	18.91	14.4	19.5

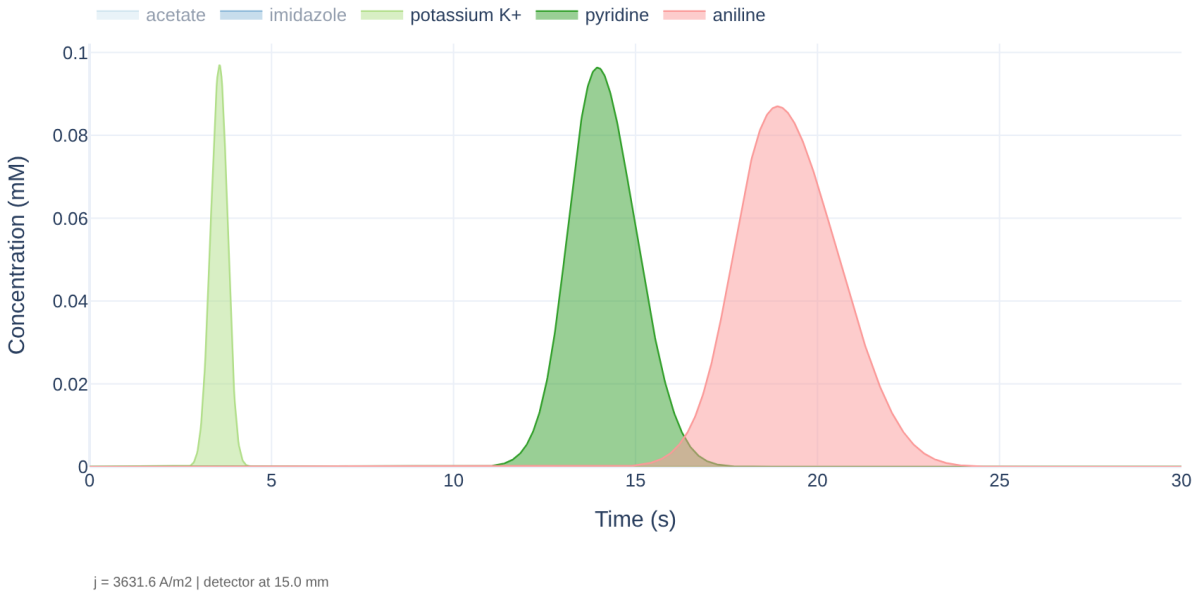


Figure 9. Pyridine and aniline at pH 4.93, 1000 V. Pyridine arrives first even though fully protonated anilinium is the faster ion.

Point to land: the order reversed from the fully protonated order. At pH 4.93 pyridine (pK_a 5.16) is 63 % protonated and aniline (pK_a 4.80) only 43 %, so pyridine carries charge for more of the time. A weak electrolyte's speed depends on its charge fraction as much as on its mobility.

On the 3 % gap: both measured apparent mobilities run about 3 % above α μ_ep. Both bases are slower than imidazolium (52.0), so both zones tail slightly (asymmetry 1.3 and 1.4 at 10 % height), and a tailing peak reaches its maximum early. This is the Part B effect, and it biases any mobility taken from the time at the maximum.

Q15 — Co-migration pH

Set α μ_ep equal:

$$\frac{30.0}{1+10^{pH-5.16}} = \frac{32.5}{1+10^{pH-4.80}} \quad \Rightarrow \quad pH = 4.00, \quad \mu_{eff} = 28.1$$

(Solve numerically, by trial pH values, or by plotting both sides.)

$$4.00 = 4.756 + \log \frac{c_{im}}{20 - c_{im}} \quad \Rightarrow \quad c_{im} = 2.98 \approx 3 \text{ mM}$$

With 3 mM imidazole the app reports **pH 4.02**, and the two peaks co-migrate almost exactly: pyridine at 8.95 s and aniline at 8.94 s. Above pH 4.00 pyridine leads, and below it aniline leads.

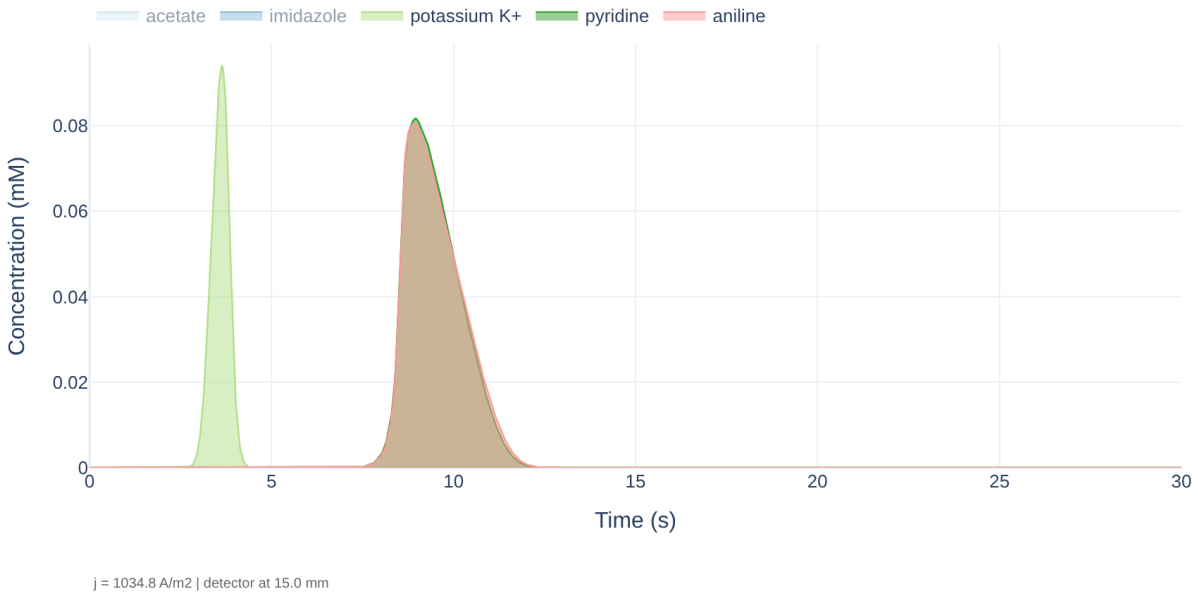


Figure 10. 3 mM imidazole (pH 4.02), 1000 V. Pyridine and aniline overlap almost perfectly at the predicted pH. Both tail strongly.

For discussion: the merged peak tails strongly (asymmetry 3.1 and 3.4). With only 3 mM imidazolium, the co-ion is dilute, so 0.1 mM of slow sample cations changes the local conductivity more than at 12 mM, which is Part B again. The tailing also pulls both apparent mobilities up to 30.3–30.4, about 8 % above the predicted 28.1. K^+ shows the mirror image: it fronts (asymmetry 0.74), and its apparent mobility reads low (74.6 against 76.2). The crossover pH is still predicted correctly because both bases are biased the same way.

An intermediate check if a group rounds to 4 mM: pH 4.16, pyridine 9.32 s and aniline 9.64 s, still in the pH 4.93 order but nearly merged.

Q16 — Class checkpoint

Any defensible choice. The strongest answers connect each effect to a preparation step: plug length or injection time (Part A), sample concentration relative to the co-ion and co-ion choice (Part B), the salt content of the sample (Part C), and the BGE pH relative to the analytes' pK_a values (Part D).

Homework

H1 — N against ℓ^2

Detector (mm)	ℓ (mm)	Run time (s)	t K^+ (s)	FWHM K^+ (s)	N K^+
10	4	20	3.18	1.09	47
15	9	20	7.22	1.13	226
20	14	30	11.26	1.13	550
25	19	36	15.19	1.13	1001

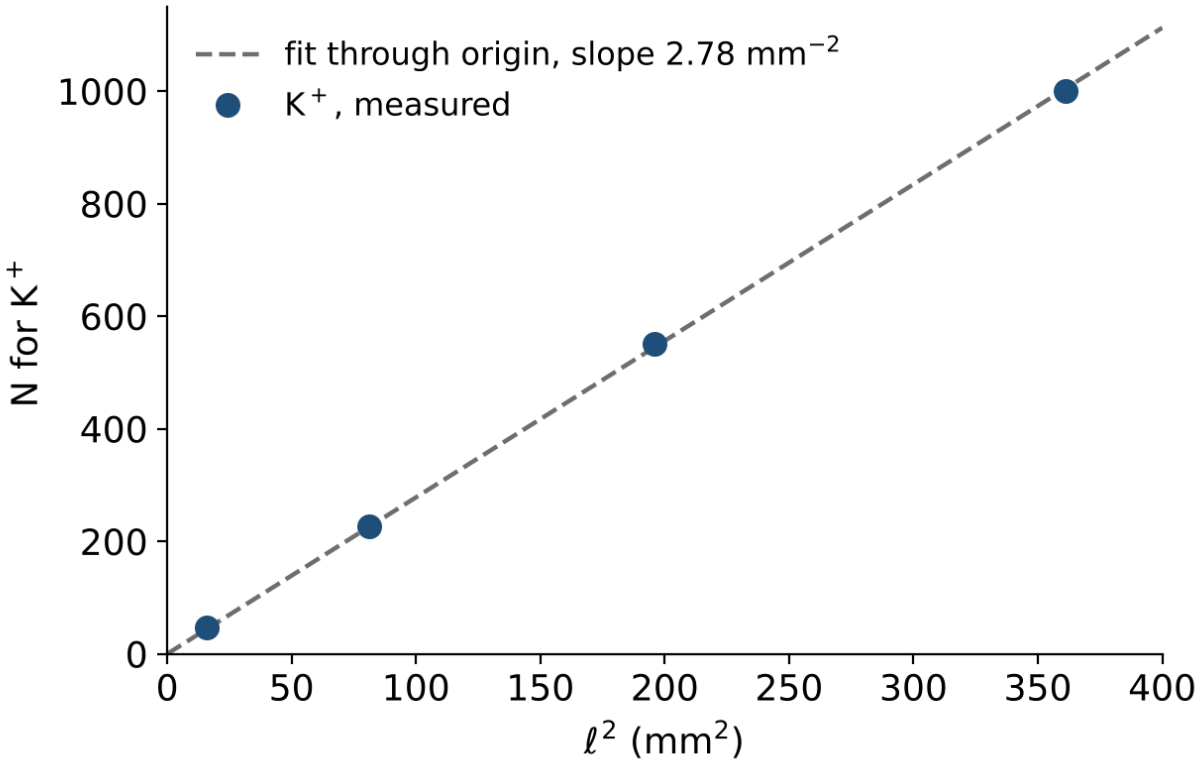


Figure 11. The plot students should produce for H1.

A least-squares line through the origin gives a slope of 2.78 mm⁻², so σ = 1/√2.78 = **0.60 mm**, against the 0.57 mm stated. The slightly larger value is the diffusion variance, which grows with ℓ and is folded into the slope.

H2 — pH selectivity for K^+ and NH_4^+

$$\alpha_{NH_4^+} = \frac{1}{1 + 10^{pH - pK_a}} \qquad \mu_{eff} = \alpha\, \mu_{ep}$$

Chloride (mM)	App pH	α	μ_eff predicted	μ_app measured	Rs at 15 mm	Rs at 25 mm
10	8.38	0.881	67.1	65.8	0.51	1.00
2	9.22	0.517	39.4	36.9	2.68	4.91

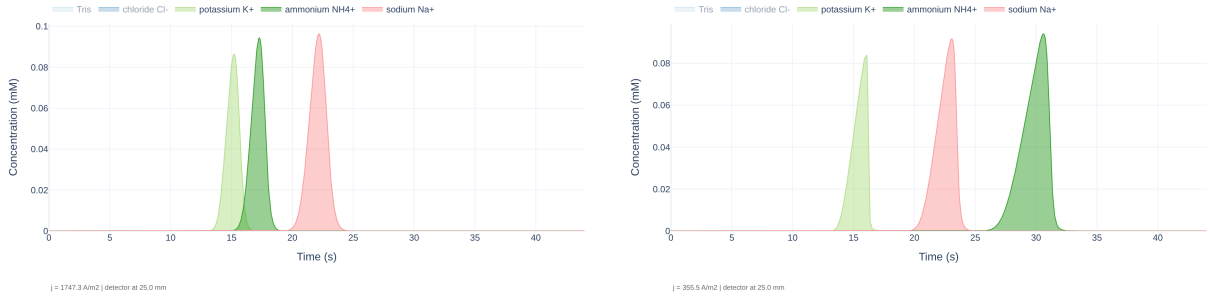


Figure 12. K^+/NH_4^+ at detector 25 mm: pH 8.38 (left, 10 mM chloride) and pH 9.22 (right, 2 mM chloride). At pH 9.22 NH_4^+ migrates behind the Na^+ marker, and all three peaks front because the co-ion ($TrisH^+$, about 2 mM) is dilute.

Measured times (K^+, NH_4^+): 10 mM chloride, 15 mm: 7.25 / 8.25 s, FWHM 1.15 / 1.18 s. At 25 mm: 15.21 / 17.28 s, FWHM 1.22 / 1.22 s. 2 mM chloride, 15 mm: 7.67 / 14.73 s, FWHM 1.20 / 1.91 s. At 25 mm: 16.04 / 30.60 s, FWHM 1.31 / 2.19 s.

Point to land: the two mobilities are equal, so all the selectivity comes from the fraction of NH_4^+ that is charged, and pH sets it. Moving the detector also doubles Rs at pH 8.38, which ties back to Part A.

K^+ shape at 2 mM chloride: the K^+ peak fronts strongly (asymmetry 0.40 at 15 mm, 0.14 at 25 mm), even though K^+ is only 0.1 mM. Only about 2 mM of Tris is protonated (it balances 2 mM chloride), so the co-ion is dilute, and 0.1 mM K^+ is 5 % of it. K^+ is faster than $TrisH^+$, so the zone fronts. This is Part B's electromigration dispersion returning because the BGE became dilute, not because the sample became concentrated.

H3 — Joule heating

0.41 W/m at 1500 V (500 V/cm) in the app's fixed 50 μm bore, radial ΔT 0.05 K. That is well under 1 W/m, so heating had no visible effect.

$$\frac{P}{L} = \kappa E^2 \pi r^2$$

Doubling the field quadruples P/L, doubling the bore radius quadruples it, and doubling κ doubles it. Accept any two with the correct exponent. The app's bore is fixed, so the field and the BGE concentration are the two a student can actually change.